

Surrogates 7020

Chapter 5(Part 2): Gaussian Process Regression

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Bayesian Inference

A Bayesian version additionally requires priors on unknown parameters, β, τ^2, g, θ . The data generated mechanism is assumed:

$$\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}_n\beta, \tau^2 \mathbf{K}_n), \mathbf{K}_n = \mathbf{C}_n + g\mathbb{I}_n,$$

Likelihood function is:

$$L(\beta, \tau^2, \theta, g) = (2\pi\tau^2)^{-\frac{n}{2}} |\mathbf{K}_n|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2\tau^2}(\mathbf{Y}_n - \mathbf{X}_n\beta)\mathbf{K}_n^{-1}(\mathbf{Y}_n - \mathbf{X}_n\beta)\right\}. \quad (1)$$

We've already seen how to infer τ^2, θ and g hyperparameters – no priors required. Conditional on $\hat{\theta}$ and $\hat{g}$ -values, defining $\mathbf{K}_n$, the same MLE calculations for $\hat{\beta}$ from your first stats class are easy with calculus. See exercises in 5.5 (book). There's really nothing Bayesian about it until you start putting priors on all unknowns: θ, g, τ^2 , and β . Right? (See Book for a discussion)

Bayesian Inference

But the fact that we can do it, i.e., force a Bayesian GP interpretation, begs three questions.

- ▶ What is the prior over?
- ▶ What likelihood is being paired with that prior (if not the MVN we've been using, which doesn't need a prior to work magic)?
- ▶ What useful “thing” do we extract from this Bayesian enterprise that wasn't there before?

The answers to those questions lie in the concept of a latent random field.

Latent random field

Another apparatus that machine learners like is graphical models, or diagrammatic depictions of Bayesian hierarchical models. The prevailing such representation for GPs is similar to schematics used to represent hidden Markov models (HMMs).

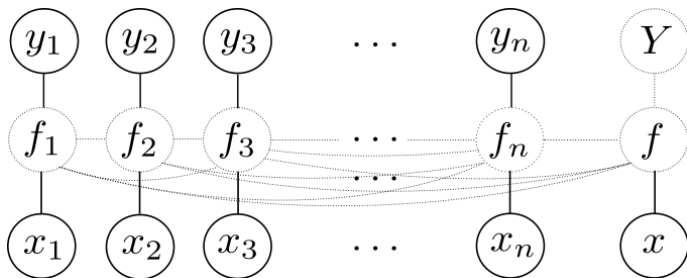


Figure: Diagram of the latent random field; similar to one from Rasmussen and Williams (2006).

Latent random field

The Bayesian interpretation inherited by the posterior predictive distribution hinges on placing a GP prior on latent fi-values, not directly on measured y_i 's.

$$F \sim \mathcal{N}_n(0, \tau^2 \mathbf{C}_n)$$

Above I'm using $\mathbf{C}_n$ rather than $\mathbf{K}_n$ which is our notation for a correlation matrix without nugget. But otherwise this uses our preferred setup from earlier. Since there are no data ($\mathbf{Y}_n$) values in , we can be (more) comfortable about calling this a prior. It's a prior over latent functions.

Now here's how the likelihood comes in, so that we'll have all of the ingredients comprising a posterior: likelihood + prior. Take an iid Gaussian sampling model for $\mathbf{Y}_n = (y_1, \dots, y_n)$ around F :

$$\mathbf{Y} \sim \mathcal{N}_n(\mathbf{F}, \sigma^2 \mathbb{I}_n).$$

To map to our earlier notation with a nugget, choose $\sigma^2 = g\tau^2$. Embellishments may include adding a linear component $\mathbf{X}_n\beta$

Matern Covariance

If a Gaussian kernel isn't working well, or if you want finer control on mean-square differentiability, then consider a Matern family member, with

$$k_{\vartheta}(r) = \frac{2^{1-\vartheta}}{\Gamma(\vartheta)} \left(r \sqrt{\frac{2\vartheta}{\theta}}\right)^{\vartheta} K_{\vartheta}\left(r \sqrt{\frac{2\vartheta}{\theta}}\right).$$

Above K_{ϑ} is a modified Bessel function of the second kind, ϑ controls smoothness, and θ is a lengthscale as before. As $\vartheta \rightarrow \infty$, i.e., a very smooth parameterization, we get the square exponential.

Matern Function

```
matern <- function(r, nu, theta) {  
  rat <- r*sqrt(2*nu/theta)  
  C <- (2^(1 - nu))/gamma(nu) * rat^nu * besselK(rat, nu)  
  C[is.nan(C)] <- 1  
  return(C)  
}  
  
r <- seq(eps, 3, length=100)  
plot(r, matern(r, nu=1/2, theta=1), type="l", ylab="k(r,nu)")  
lines(r, matern(r, nu=2, theta=1), lty=2, col=2)  
lines(r, matern(r, nu=10, theta=1), lty=3, col=3)  
legend("topright", c("nu=1/2", "n=2", "nu=10"), lty=1:3, col=1:3, bty="n")
```

Smoothness

```
X <- seq(0, 10, length=100)
R <- sqrt(distance(X))
K0.5 <- matern(R, nu=1/2, theta=1)
K2 <- matern(R, nu=2, theta=1)
K10 <- matern(R, nu=10, theta=1) + diag(eps, 100)

par(mfrow=c(1,3))
matplot(X, t(rmvnorm(3,sigma=K0.5)), type="l", col=1, lty=1,
  xlab="x", ylab="y, nu=1/2")
matplot(X, t(rmvnorm(3,sigma=K2)), type="l", col=2, lty=2,
  xlab="x", ylab="y, nu=2")
matplot(X, t(rmvnorm(3,sigma=K10)), type="l", col=3, lty=3, xlab="x",
  ylab="y, nu=10")
```


Differentiable

```
F <- rmvnorm(1, sigma=K2)
dF <- (F[-1] - F[-length(F)])/(X[2] - X[1])
d2F <- (dF[-1] - dF[-length(dF)])/(X[2] - X[1])

plot(X, F, type="l", lwd=2, xlim=c(0,13), ylim=c(-5,5),
     ylab="F and derivatives")
lines(X[-1], dF, col=2, lty=2)
lines(X[-(1:2)], d2F, col=3, lty=3)
legend("topright", c("F", "dF", "d2F"), lty=1:3, col=1:3, bty="n")
```